

## MSc Thesis Proposal

# Bridging the Gap: Cross-Platform Domain Adaptation for Rumex Weed Detection

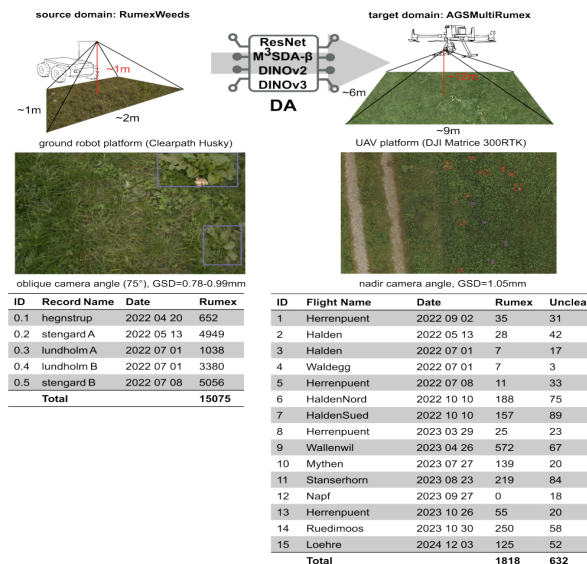
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## Motivation

Precision weed control relies on both ground robots and UAVs, but every new sensing platform typically starts from scratch on labelled data; one of the biggest bottlenecks in agricultural computer vision. RumexWeeds provides thousands of bounding-box labels for Rumex obtusifolius collected by a ground robot, while labelled drone imagery for the same weed remains scarce. Recent work [2] showed that a classifier trained on RumexWeeds collapses on drone imagery unless adapted, but recovers when built on large pretrained backbones. This project asks a more practically relevant question: can an object detector, not just a classifier, be made to transfer robustly from the ground-robot domain to the drone domain, using domain adaptation techniques rather than simply scaling up the backbone?

**Keywords:** Cross-Domain Generalization · Domain-Adaptive Object Detection · Weed Detection · UAV · Precision Agriculture



## Problem Statement

RumexWeeds (source domain) and the drone dataset AGSMultiRumex (target domain) [2] depict the same weed species but under a substantially different setup: RumexWeeds is captured at 1 m height, 75° angle, close range from a ground robot, while AGSMultiRumex is nadir UAV imagery at 12 m altitude with very different scale, background clutter, and vegetation composition. Prior work showed this gap nearly breaks standard CNN classifiers and is only partly closed by moment-matching or by scaling up self-supervised backbones. This project treats the drone domain as (mostly) unlabelled at training time, matching a realistic deployment scenario where labelling a new sensing platform from scratch is expensive.

## Proposed Approach

The student will work with the RumexWeeds dataset, using it to train a modern object detector for Rumex obtusifolius on ground-robot imagery, and will then evaluate how well this detector transfers to the drone-based AGSMultiRumex dataset introduced in [2], which depicts the same weed under a different viewpoint, altitude, and imaging setup. Building on the domain gap identified in that work, the student will explore how to close it at the detection level, rather than relying on scaling up the backbone alone, and will analyse performance across the individual drone flights to understand which conditions (lighting, growth stage, confounding species) remain hardest to generalize to.

## References to Get Started

- [1] [RumexWeeds: A Grassland Dataset for Agricultural Robotics](#)
- [2] [Towards Robust Deep Learning-based Rumex Obtusifolius Detection from Drone Images](#)
- [3] [Domain Adaptive Faster R-CNN for Object Detection in the Wild](#)
- [4] [Large Self-Supervised Models Bridge the Gap in Domain Adaptive Object Detection](#)